MIL-F-2763B
1 December 1966
SUPERSEDING
MIL-F-2763A
15 May 1961

MILITARY SPECIFICATION

FILTER, LIGHT, OPTICAL INSTRUMENT: BINOCULAR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers the minimum essential requirements for one type of variable density filter (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-E-760	Ethyl Alcohol (Ethanol)
PPP-T-76	Tape, Pressure Sensitive, Adhesive, Paper, Water Resistant
PPP-B-585	Box, Wood, Wirebound
PPP-B-591	Box, Fiberboard, Wood-Cleated
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-P-13988	Paper, Lens, Tissue, Antitarnish, Wrapping

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-1241	Optical Terms and Definitions
MS 51600	Filter, Light, Optical Instrument

FSC 6650

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DRAWINGS

U,S. Army Munitions Command

D7691956 Filter Ml Assembled and Sectional Views

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Construction.- The design, material, and dimensions of the filter shall be in accordance with MS 51600 and Drawing D7691956 which form a part of this specification.

3.2 Light transmission.- The transmission of the assembled filters for normally incident light through a wave length range of 400 millimicrons to 700 millimicrons shall be as follows:

(a) Transmission for white light with the polarizing axis parallel 8, plus or minus 4 percent.

(b) Maximum transmission for white light with the polarizing axis crossed 1.0 percent.

3.3 Resolving power.- The resolving power of the assembled filter shall be 1 minute or better in all positions.

3.4 Lens power.- The positive or negative refractive power of a pair of polarizing disks shall not exceed 1/4 diopter in any meridian. The difference in refractive power in any two meridians shall not exceed 1/4 diopter.

3.5 Deviation.- Two parallel rays incident to the two pairs of polarizing disks shall emerge parallel within 10 minutes of arc irrespective of density setting.

3.6 Temperature requirements.

3.6.1 Operation.- After being temperature stabilized at ambient temperatures of -40° and +150°F and while still at these temperatures the filter rotating mechanism shall be mechanically operable, with or without gloves.

3.6.2 Storage.- After being temperature stabilized at ambient temperatures of -80° and +160°F the filter shall be stabilized at room temperature and shall show no evidence of discoloration, separation, bubbles, or other damage.

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3.7 Identification.- The filter shall be inscribed with the following identification in accordance with MIL-STD-130.

V. D. Filter
MS 51600-1

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Optical terms and definitions.- Terms and definitions used in this specification, which are peculiar to the general field of optics, are defined in MIL-STD-1241.

4.2 Quality conformance inspections.

4.2.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.2.2 Inspection system.- Except as otherwise provided for by the contract, the contractor shall provide and maintain an inspection system in accordance with the applicable requirements of MIL-I-45208. (See 6.2).

4.2.3 Classification of defects.- The classification of defects in Tables I and II shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots.

TABLE I - Classification of Defects

Use Inspection Level II, and Sampling Plan Table IV-A
of MIL-STD-105

Critical: None defined

Major: AQL 2.5%	Requirement Paragraph	QA. Paragraph
101. Construction	3.1	4.3.1
102. Light transmission	3.2	4.3.2
103. Resolving power	3.3	4.3.3

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TABLE I - Classification of Defects (Cont'd)

Major: AQL 2.5%	Requirement Paragraph	QA. Paragraph
104. Lens power	3.4	4.3.4
105. Deviation	3.5	4.3.5
106. Identification	3.7	4.4

TABLE II - Classification of Defects

Use Inspection Level S4, and Sampling Plan Table IV-A of MIL-STD-105

Critical: None defined

Major: AQL 6.5%	Requirement Paragraph	QA. Paragraph
101. Operation	3.6.1	4.3.6.1
102. Storage	3.6.2	4.3.6.2

4.3 Test methods.

4.3.1 Construction.- The construction of the filters shall be inspected visually and by means of standard measuring equipment for compliance with the requirements of 3.1.

4.3.2 Light transmission.- Light transmission for normally incident light shall be measured by means of an approved photometric method with the polarizing axes crossed and with the polarizing axes parallel. Percent transmission shall not exceed the limits specified in 3.2.

4.3.3 Resolving power.- The filters shall be tested for resolving power utilizing a National Bureau of Standards Test Chart, located at the required distance for resolving the requirements of 3.3. A three-power auxiliary telescope may be used to assist the observer's eye. The test pattern shall be resolved in each of the four meridians and shall meet the requirements specified in 3.3.

4.3.4 Lens power.- The positive or negative refractive power of the polarizing disks shall be measured by placing the filter in front of a dioptometer of at least 3-power and sighting on a calibrated target. The difference in refractive power shall not exceed the tolerance specified in 3.4 in any meridian at any density setting.

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4.3.5 Deviation.- Deviation of the two parallel rays with respect to each other shall be measured by viewing an effectively remote target through a special testing device having a pair of auxiliary telescopes with suitably calibrated reticles. The deviation of the polarizing rays shall be measured by placing the disks in front of the telescopes' objectives and noting the relative displacement of the image. Displacement shall not exceed the tolerance specified in 3.5 at any density setting.

4.3.6 Temperatures.

4.3.6.1 Operation.- The operation temperature test shall consist of exposing the filters to the temperatures indicated in 3.6.1 by means of a cold chest and an oven capable of maintaining these temperatures until the units under test are thermally stabilized. At the stabilized temperatures, the filters shall be tested to determine compliance with the requirements specified in 3.6.1.

4.3.6.2 Storage.- The filters shall be exposed to the temperatures indicated in 3.6.2 by means of a cold chest and an oven capable of maintaining these temperatures until the units under test are thermally stabilized. The filters shall then be thermally stabilized at room temperature at which time the filters shall be visually examined to determine compliance with the requirements specified in 3.6.2.

4.4 Identification.- The identification marking shall be visually inspected for compliance with MIL-STD-130 and the requirements of 3.7.

4.5 Preservation and packaging tests.- The test procedures in MIL-P-116 shall be used for the following packaging and preservation tests to determine the minimum acceptable quality.

4.5.1 Sampling for test.- Samples for the tests listed in Table IV shall be selected from lots that have met the minimum inspection requirements listed in Table III.

TABLE III - Classification of Defects

Use Inspection Level II, and Sampling Plan Table IV-A of MIL-STD-105

Critical: None defined

Major: AQL 2.5%	Requirement Paragraph
101. Improper cleaning (Level A)	5.1.1.1
102. Incorrect packaging method (Level A)	5.1.1.2
103. Intermediate packaging not as specified (Level A)	5.1.1.3
104. Marking illegible, incorrect, not as specified (Levels A, B, C)	5.3

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TABLE III - Classification of Defects (Cont'd)

Minor: AQL 4.0%	Requirement Paragraph
201. Gross weight exceeded (Levels A, B)	5.2.1, 5.2.2
202. Shipping containers not as specified (Levels A, B)	5.2.1, 5.2.2

TABLE IV - Classification of Defects

Use Inspection Level S4, and Sampling Plan Table IV-A of MIL-STD-105

Critical: None defined

Major: AQL 4.0%	Requirement Paragraph
101. Heat seal (Level A)	5.1.1.2
102. Quick leak (Level A)	5.1.1.2

4.5.2 Level "C" preservation, packaging and packing.- Due to the variance of preservation, packaging and packing that would conform to this level of preservation, packaging and packing, the contractor must submit his intended preservation, packaging and packing methods and inspection procedures to the procuring activity for approval.

4.5.3 Level "C" shipping container.- Filters packaged as specified in 5.1.2 shall be packed to the applicable regulation for the mode of transportation utilized in compliance with 5.2.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning.- The lens of the filter shall be cleaned with alcohol conforming to O-E-760 and wrapped in lens tissue paper conforming to MIL-P-13988.

5.1.1.2 Unit packaging.- Each filter shall be individually packaged Method IA-8 of MIL-P-116.

5.1.1.3 Intermediate packaging.- Twenty unit packages shall be further packaged in a close fitting fiberboard box conforming to class 2, W5c of PPP-B-636. The box shall be closed by applying at least a 2 inch width tape conforming to PPP-T-76 to top and bottom seams. If bottom of the container is closed with staples or adhesive only a single piece of tape on the top seam is required.

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5.1.2 Level C.- The filters shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.- Packing shall be levels A, B or C as specified (see 6.2).

5.2.1 Level A.- Filters packaged as specified in 5.1.1 shall be packed in a wirebound wood box, wood-cleated fiberboard box, wood-cleated plywood box, nailed wood box, or a fiberboard box conforming to PPP-B-585, class 3; PPP-B-591 (Overseas type); PPP-B-601, style A or B; PPP-B-621, class 2; PPP-B-636, class 2, respectively. Closure and strapping shall be as specified in the appendix of the applicable box specification. The gross weight shall not exceed 200 pounds.

5.2.2 Level B.- Filters packaged as specified in 5.1.1 shall be packed in a wirebound wood box, wood-cleated fiberboard box, wood-cleated plywood box, nailed wood box or a fiberboard box conforming to PPP-B-585, class 1; PPP-B-591 (Domestic type); PPP-B-601 (Domestic type); PPP-B-621, class 1; and PPP-B-636, class 2, respectively. The gross weight of the boxes shall not exceed the weight limitation of the applicable box specification. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C.- Unless otherwise specified by the contracting officer filters shall be packed in a manner that will insure safe transportation at the lowest cost to the point of delivery.

5.3 Marking.- In addition to any special marking required by the contract or order, unit packages, intermediate packages and shipping containers shall be marked in accordance with the applicable requirements of MIL-STD-129.

6. NOTES

6.1 Intended use.- Filters covered by this specification control or moderate the intensity of light transmitted by rotating the polarizing disks. The filters are designed to spring fit over annular grooves around the periphery of eyepiece guards of binoculars.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing required.
(c) Administration of inspection system (4.2.2).

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Custodians:
Army - MU
Navy - SH
Air Force - 82

Review activities:
Army - MU
Navy -
Air Force - 82

User activities:
Army - MI
Navy - SH, OS, MC
Air Force -

Preparing activity:
Army - MU

Project No. 6650-0169

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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